

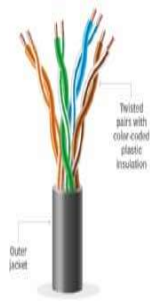
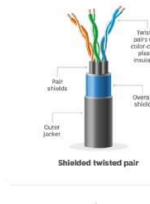
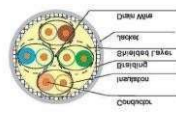
Practical-2

Aim: Study of different types of Network cables.

a) Understand different types of network cable.

Different type of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial Cable
4. Fibre Optic Cable

Cable type	Category	Maximum Data Transmission	Advantages/ Disadvantages	Application/Use	Image
UTP	Category 3	10 bps	<u>Advantages</u> • Cheaper in cost • Easy to install as they have a smaller overall diameter. <u>Disadvantages</u> • More prone to (EMI) Electromagnetic interference and noise	10Base-T Ethernet	
	Category 5	Up to 100 Mbps		Fast Ethernet, Gigabit Ethernet	
	Category 5e	1Gbps		Fast Ethernet, Gigabit Ethernet	
STP	Category 6, 6a	10Gbps	<u>Advantages</u> • Shielded. • Faster than UTP. • Less susceptible to noise and interference <u>Disadvantages</u> • Expensive • Greater installation effort	Gigabit Ethernet, 10G Ethernet (55m) Widely used in data centres	
SSTP	Category 7	10Gbps		Gigabit Ethernet, 10G Ethernet (100m)	 

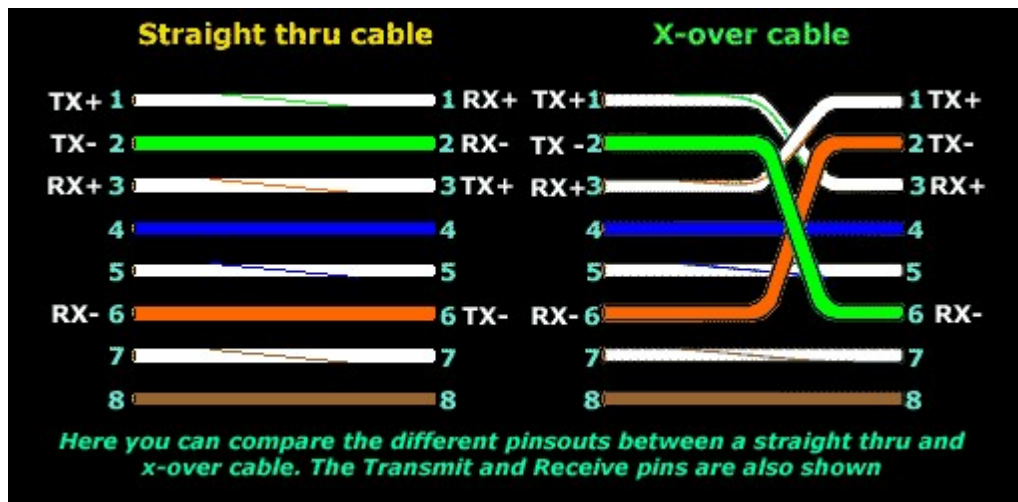
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Coaxial cable	RG-6 RG-59 RG-11	10-100Mbps	• High bandwidth • Immune to interference • Low loss bandwidth • Versatile • Disadvantages • Limited distance • Cost • Size is bulky	Speed of signal is 500m Television network High speed internet connections	
fibre optics cable	Single mode Multi mode	100Gbps	Advantages • High speed • High bandwidth • High security • Long distance Disadvantages • Expensive • Requires skilled installers	• Maximum distance of fibre optics cable is around 100meters	

b) Make Your Own Ethernet Cross-Over Cable/ Straight cable

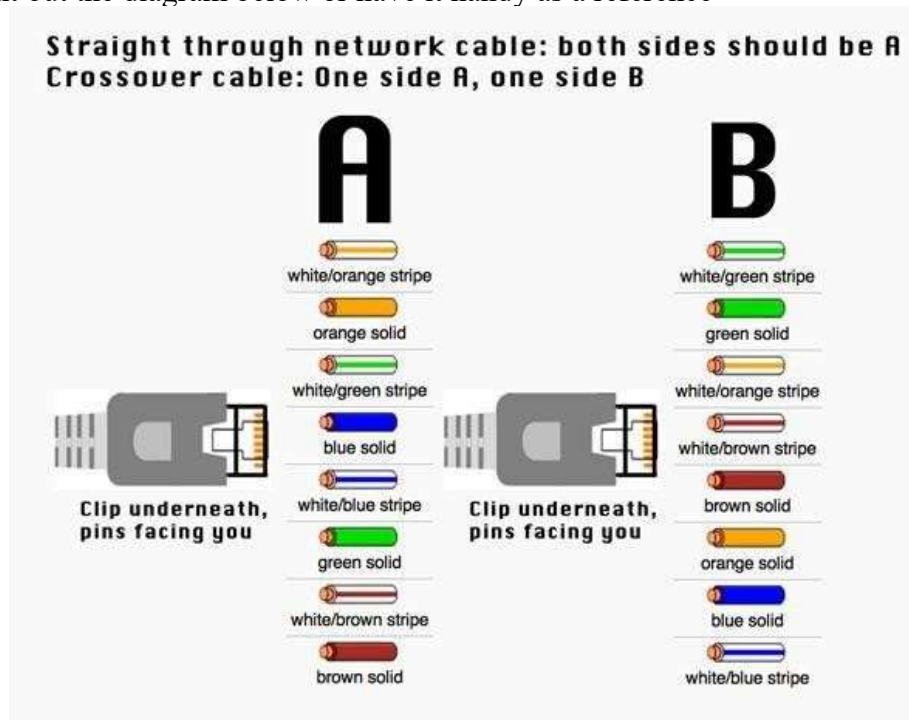
Tools and parts needed:

- Ethernet cabling. CAT5e is certified for gigabit support, but CAT5 cabling works as well, just over shorter distances.
- A crimping tool. This is an all-in-one networking tool shaped to push down the pins in the plug and strip and cut the shielding off the cables.
- Two RJ45 plugs.
- Optional two plug shields.



Difference between crossover cable and straight cable

Take a print out the diagram below or have it handy as a reference

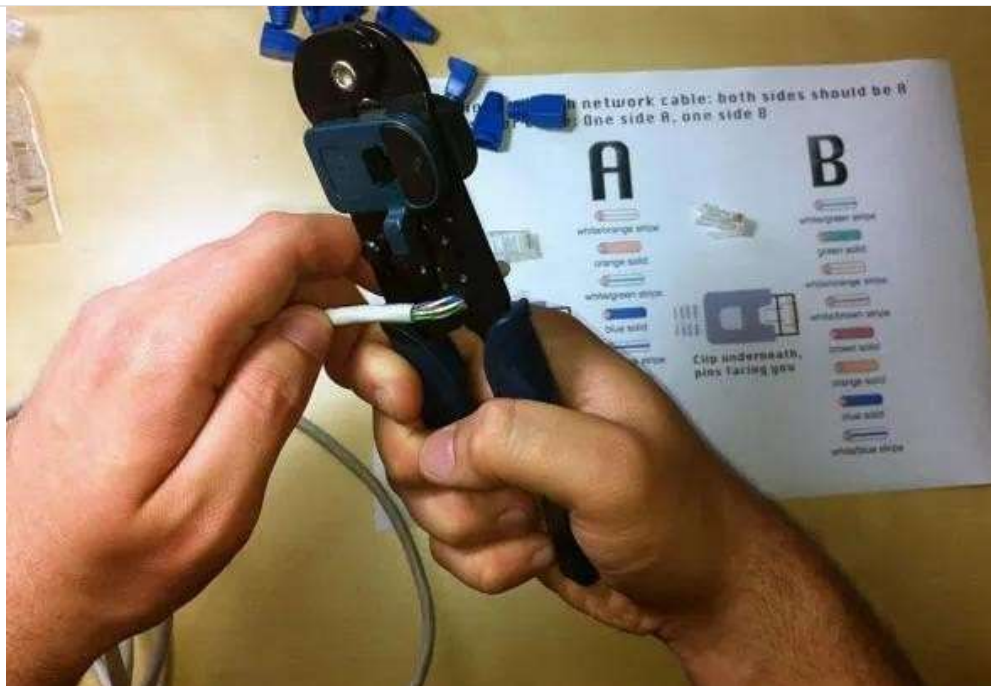


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Step 1: To start construction of the device, begin by threading shields onto the cable.

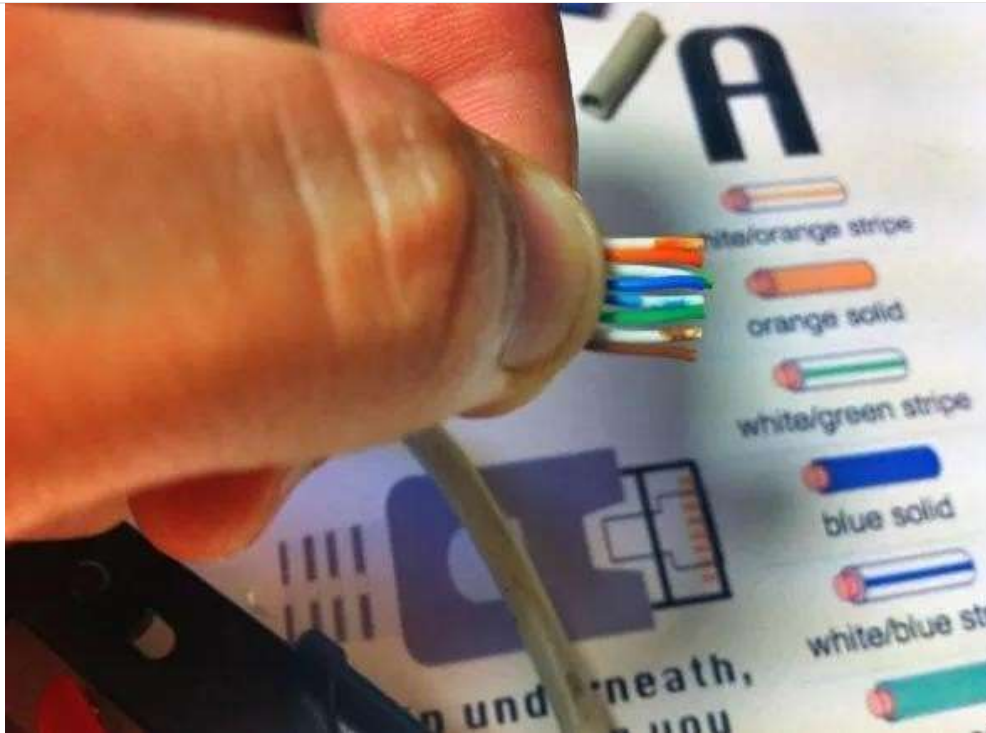


Step 2: Next, strip approximately 1.5 cm of cable shielding from both ends. The crimping tool has a round area to complete this task.

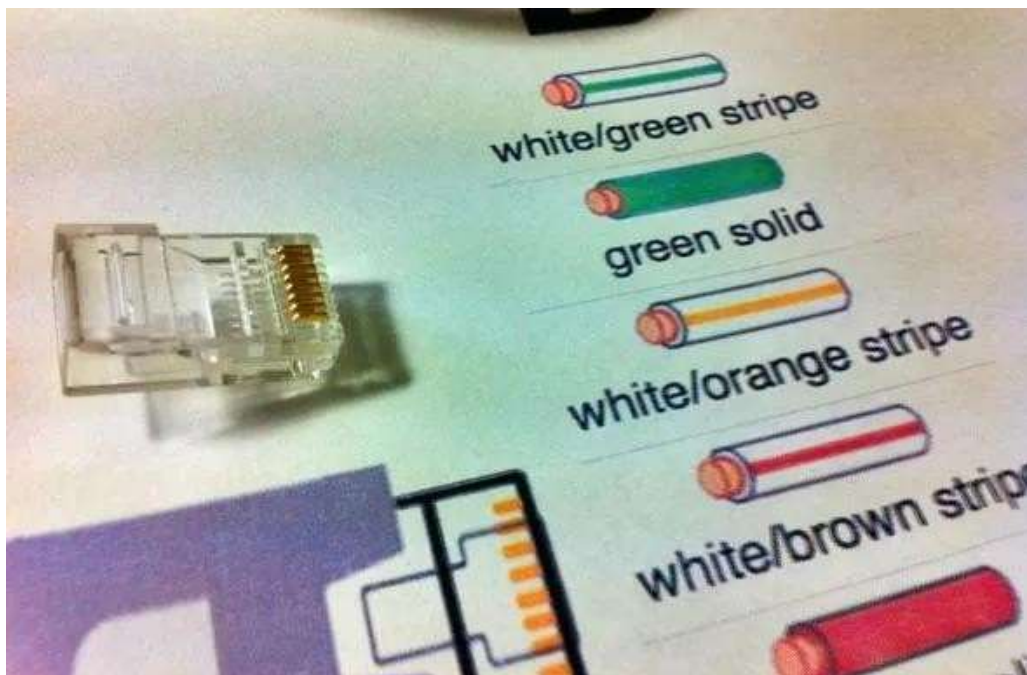


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Step 3: After, you will need to untangle the wires; there should be four “twisted pairs.” Referencing back to the sheet, arrange them from top to bottom. One end should be in arrangement A and the other in B.



Step 4: Once the order is correct, bunch them together in a line, and if there are any that stick out farther than others, snip them back to create an even level. The difficult aspect is placing these into the RJ45 plug without messing up the order. To do so, hold the plug with the clip side facing away from you and have the gold pins facing toward you, as shown.



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Step 5: Next, push the cable right in. The notch at the end of the plug needs to be just over the cable shielding, and if it isn't, that means that you stripped off too much shielding. Simply snip the cables back a little more.



Step 6: After the wires are securely sitting inside the plug, insert it into the crimping tool and push down.

It should be shaped correctly, but pushing too hard can crack the fragile plastic plug.

Step 7: Lastly, repeat for the other end using diagram B (to make a crossover cables)/ using diagram A (to make a straight through cable)

To test it, plug it in and attempt to connect two devices directly.

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Network Cables:

This section provides additional information about commonly used networking cables beyond the basic cables covered in the lab manual. It can be included as supplementary study material for GitHub documentation.

Cat6 Cable

Supports up to 10 Gbps over short distances (55 m). It has improved insulation and reduced crosstalk compared to Cat5e. Commonly used in modern office and campus networks.

Cat6a Cable

Supports 10 Gbps up to 100 m. It offers better shielding and performance than Cat6 and is suitable for enterprise networks and data centers.

Cat7 Cable

Shielded twisted pair cable designed for very high-speed networks. Supports frequencies up to 600 MHz and provides excellent protection against electromagnetic interference.

Cat8 Cable

Supports 25–40 Gbps over distances up to 30 m. Primarily used in data centers and high-performance server connections.

Twinax Cable

A short-distance copper cable used mainly for connecting switches, routers, and servers in data centers. Available as passive and active direct-attach cables.

Ribbon Fiber Cable

Contains multiple optical fibers arranged in parallel. It is used in high-density fiber installations and backbone network infrastructure.

Armored Fiber Cable

A fiber optic cable with a protective metal layer that provides resistance against physical damage, rodents, and harsh outdoor environments.

Direct Burial Cable

Designed for underground installation without additional conduit protection. Used in campus networks and outdoor communication links.

Patch Cable

A short Ethernet cable used to connect computers, switches, routers, and patch panels. Available in Cat5e, Cat6, Cat6a, and Cat8 variants.

Rollover (Console) Cable

Used for configuring network devices such as Cisco routers and switches through the console port. It is not used for normal Ethernet data transmission.

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Comparison of Additional Cables:

Cable	Speed	Distance	Typical Use
Cat6	10 Gbps	55 m	Office LANs
Cat6a	10 Gbps	100 m	Enterprise networks
Cat7	10 Gbps	100 m	Shielded installations
Cat8	25–40 Gbps	30 m	Data centers
Twinax	10–100 Gbps	1–7 m	Server connections
Ribbon Fiber	100+ Gbps	Long distance	Backbone networks
Armored Fiber	100+ Gbps	Long distance	Outdoor networks

Conclusion

The additional cable types discussed above are widely used in modern networking environments. Understanding their speed, distance limitations, shielding, and applications helps in selecting the appropriate cable for LANs, data centers, backbone networks, and outdoor communication systems.